

Greedy Construction Baseline: Heuristic Formulation and Complexity Analysis

Quantum Annealing-Based Rolling Stock Scheduling Research Suite

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Abstract

This document details the deterministic Greedy Construction heuristic for the Rolling Stock Scheduling and Circulation Problem. We present the formal algorithmic procedure in pseudocode using `algorithmicx`, analyze its asymptotic time complexity of $\mathcal{O}(|T| \cdot |U|)$, and theoretically examine why greedy dispatch fails due to myopic decision-making and lack of downstream look-ahead.

1 Introduction

The greedy construction algorithm establishes a deterministic, sequential baseline for the rolling stock circulation problem. Timetable trips are processed strictly in chronological order of scheduled departure:

$$T = (i_1, i_2, \dots, i_{|T|}), \quad \text{such that } \text{dep}_{i_1} \leq \text{dep}_{i_2} \leq \dots \leq \text{dep}_{i_{|T|}} \quad (1)$$

At each step, the algorithm assigns trainset units to the current trip based on immediate temporal availability and minimum deadhead repositioning distance.

2 Unit State Representation

For each rolling stock unit $u \in U$, a dynamic operational state is maintained:

$$\sigma_u = (\text{loc}_u, \text{avail}_u, \mu_u) \quad (2)$$

initialized at $t = 0$ as:

$$\text{loc}_u^{(0)} = h_u \quad (\text{initial home depot}) \quad (3)$$

$$\text{avail}_u^{(0)} = a_u \quad (\text{earliest departure time}) \quad (4)$$

$$\mu_u^{(0)} = \mu_u^{\text{init}} \quad (\text{initial accumulated mileage}) \quad (5)$$

3 Algorithmic Procedure

Algorithm 1 presents the complete pseudocode for the greedy construction heuristic.

Algorithm 1 Greedy Construction Heuristic for Rolling Stock Circulation

Require: Fleet U , Timetable trips T , Distance matrix δ , Travel time matrix t , Demands ρ

Ensure: Circulation chains $\mathcal{C}_u$ for all $u \in U$, Schedule $\mathcal{S}$

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1: Sort trips  $T = (i_1, i_2, \dots, i_{|T|})$  such that  $\text{dep}_{i_1} \leq \text{dep}_{i_2} \leq \dots \leq \text{dep}_{i_{|T|}}$ 
2: for each unit  $u \in U$  do
3:    $\mathcal{C}_u \leftarrow \emptyset$ ,  $\text{loc}_u \leftarrow h_u$ ,  $\text{avail}_u \leftarrow a_u$ ,  $\mu_u \leftarrow \mu_u^{\text{init}}$ 
4: end for
5: for each trip  $i \in T$  do
6:    $\text{unmet\_demand} \leftarrow \rho_i$ 
7:    $\mathcal{U}_{\text{feas}} \leftarrow \emptyset$ 
8:   for each unit  $u \in U$  do
9:      $\text{transit\_time} \leftarrow t(\text{loc}_u, o_i)$ 
10:    if  $\text{avail}_u + \text{transit\_time} \leq \text{dep}_i$  then
11:       $\text{dist} \leftarrow \delta(\text{loc}_u, o_i)$ 
12:      Add  $(u, \text{dist})$  to  $\mathcal{U}_{\text{feas}}$ 
13:    end if
14:  end for
15:  if  $\mathcal{U}_{\text{feas}} \neq \emptyset$  then
16:    Sort  $\mathcal{U}_{\text{feas}}$  in ascending order of deadhead distance  $\text{dist}$ 
17:    Select best primary unit  $u^* \leftarrow \arg \min_{(u,d) \in \mathcal{U}_{\text{feas}}} d$ 
18:    Assign  $u^*$  to trip  $i$ :  $\mathcal{C}_{u^*} \leftarrow \mathcal{C}_{u^*} \cup \{i\}$ 
19:    Update:  $\text{loc}_{u^*} \leftarrow d_i$ ,  $\text{avail}_{u^*} \leftarrow \text{arr}_i + \tau_i$ ,  $\mu_{u^*} \leftarrow \mu_{u^*} + \delta(\text{loc}_{u^*}, o_i) + \ell_i$ 
20:     $\text{unmet\_demand} \leftarrow \max(0, \text{unmet\_demand} - c_{u^*})$ 
21:    if  $\text{unmet\_demand} > 0$  and  $\mathcal{U}_{\text{feas}} \setminus \{u^*\} \neq \emptyset$  then
22:      Select secondary unit  $u_2^* \leftarrow \arg \min_{(u,d) \in \mathcal{U}_{\text{feas}} \setminus \{u^*\}} d$ 
23:      Assign  $u_2^*$  to trip  $i$ :  $\mathcal{C}_{u_2^*} \leftarrow \mathcal{C}_{u_2^*} \cup \{i\}$ 
24:      Update:  $\text{loc}_{u_2^*} \leftarrow d_i$ ,  $\text{avail}_{u_2^*} \leftarrow \text{arr}_i + \tau_i$ ,  $\mu_{u_2^*} \leftarrow \mu_{u_2^*} + \delta(\text{loc}_{u_2^*}, o_i) + \ell_i$ 
25:    end if
26:  else
27:     $\triangleright$  Fallback: assign earliest available unit despite timing violation
28:     $u_{\text{fallback}} \leftarrow \arg \min_{u \in U} \text{avail}_u$ 
29:    Assign  $u_{\text{fallback}}$  to trip  $i$ :  $\mathcal{C}_{u_{\text{fallback}}} \leftarrow \mathcal{C}_{u_{\text{fallback}}} \cup \{i\}$ 
30:    Update state of  $u_{\text{fallback}}$  accordingly
31:  end if
32: end for
33: return circulation chains  $\{\mathcal{C}_u \mid u \in U\}$ 
```

4 Complexity Analysis

Let $|T|$ denote the total number of timetable trips and $|U|$ the total fleet size.

1. **Initial Sorting:** Sorting timetable trips chronologically by departure timestamp dep_i takes $\mathcal{O}(|T| \log |T|)$ operations.
2. **Per-Trip Evaluation Loop:** For each trip $i \in \{1, \dots, |T|\}$:
 - Evaluating temporal availability and repositioning distance across all candidate units takes $\mathcal{O}(|U|)$ checks.
 - Sorting the candidate subset $\mathcal{U}_{\text{feas}}$ takes $\mathcal{O}(|U| \log |U|)$ in the worst case, or $\mathcal{O}(|U|)$ using a single linear scan to find the minimum.
 - Updating unit state takes $\mathcal{O}(1)$.

3. **Overall Time Complexity:** Summing across all $|T|$ trips, the total running time is:

$$\mathcal{O}(|T| \log |T| + |T| \cdot |U|) \quad (6)$$

In typical practical railway networks where $|U| \geq \log |T|$, the asymptotic running time simplifies directly to:

$$\mathcal{O}(|T| \cdot |U|) \quad (7)$$

4. **Space Complexity:** Maintaining unit circulation chains and state arrays requires $\mathcal{O}(|T| + |U|)$ auxiliary memory.

5 Why Greedy Fails: Myopic Decisions and No Look-Ahead

Although the greedy construction heuristic is fast and deterministic, it exhibits fundamental failure modes when applied to dense, constrained rolling stock circulation networks:

5.1 1. Myopic Local Optimization

The greedy criterion minimizes immediate deadhead repositioning $\delta(\text{loc}_u, o_i)$ for the current trip i . It evaluates decisions purely in isolation, completely blind to future timetable commitments. Consequently, a unit is often dispatched to serve a minor trip merely because it is located nearby, leaving no units available for high-demand or geographically isolated trips scheduled shortly thereafter.

5.2 2. Lack of Downstream Look-Ahead

In railway operations, units must complete round-trip circuits returning to maintenance-equipped depots. The greedy policy has zero look-ahead capability:

- Units are stranded at peripheral terminus stations without adequate turnaround windows for scheduled return revenue trips.
- When peak evening return trips begin, the fleet is geographically misallocated across the network, forcing massive, expensive deadhead transits or catastrophic schedule delays.

5.3 3. Error Cascade and Infeasibility Propagation

Unlike metaheuristics or mathematical programming, the greedy heuristic cannot backtrack or undo an assignment once made:

$$\text{Suboptimal assignment at } t_1 \implies \text{Severe resource deficit at } t_2 \implies \text{Forced timing violations at } t_3 \quad (8)$$

As demonstrated in empirical network scaling experiments, once trip density exceeds network capacity, the greedy algorithm cascades into hundreds of minutes of timing violations and heavy unmet passenger demand.